Activity 2: Analyzing Different Scenarios and Generating the Appropriate Visualization

We'll be working with the 120 years of Olympic History dataset acquired by Randi Griffin from https://www.sports-reference.com/ and made available on the GitHub repository of this book. Your assignment is to identify the top five sports based on the largest number of medals awarded in the year 2016, and then perform the following analysis:

1. Generate a plot indicating the number of medals awarded in each of the top five sports in 2016.
2. Plot a graph depicting the distribution of the age of medal winners in the top five sports in 2016.
3. Find out which national teams won the largest number of medals in the top five sports in 2016.
4. Observe the trend in the average weight of male and female athletes winning in the top five sports in 2016.

High-Level Steps

1. Download the dataset and format it as a pandas DataFrame.
2. Filter the DataFrame to only include the rows corresponding to medal winners from 2016.
3. Find out the medals awarded in 2016 for each sport.
4. List the top five sports based on the largest number of medals awarded. Filter the DataFrame one more time to only include the records for the top five sports in 2016.
5. Generate a bar plot of record counts corresponding to each of the top five sports.
6. Generate a histogram for the Age feature of all medal winners in the top five sports (2016).
7. Generate a bar plot indicating how many medals were won by each country's team in the top five sports in 2016.
8. Generate a bar plot indicating the average weight of players, categorized based on gender, winning in the top five sports in 2016.

In [ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Étape 1 : Charger le dataset
url = "https://raw.githubusercontent.com/TrainingByPackt/Interactive-Data-Vi
df = pd.read_csv(url)

# Vérifions la structure du dataset
```

```
print("Dimensions :", df.shape)
print("\nPremières lignes :")
print(df.head())
print("\nColonnes disponibles :")
print(df.columns.tolist())
print("\nInfo sur les données :")
print(df.info())
```

```
Dimensions : (271116, 15)

Premières lignes :
   ID                     Name Sex   Age  Height  Weight            Team  \
0   1                A Dijiang   M  24.0   180.0    80.0           China
1   2                 A Lamusi   M  23.0   170.0    60.0           China
2   3       Gunnar Nielsen Aaby   M  24.0     NaN     NaN         Denmark
3   4      Edgar Lindenau Aabye   M  34.0     NaN     NaN  Denmark/Sweden
4   5  Christine Jacoba Aaftink   F  21.0   185.0    82.0     Netherlands

   NOC        Games  Year  Season       City          Sport  \
0  CHN  1992 Summer  1992  Summer  Barcelona     Basketball
1  CHN  2012 Summer  2012  Summer     London           Judo
2  DEN  1920 Summer  1920  Summer  Antwerpen       Football
3  DEN  1900 Summer  1900  Summer      Paris     Tug-Of-War
4  NED  1988 Winter  1988  Winter    Calgary  Speed Skating

                             Event Medal
0       Basketball Men's Basketball   NaN
1      Judo Men's Extra-Lightweight   NaN
2           Football Men's Football   NaN
3       Tug-Of-War Men's Tug-Of-War  Gold
4  Speed Skating Women's 500 metres   NaN

Colonnes disponibles :
['ID', 'Name', 'Sex', 'Age', 'Height', 'Weight', 'Team', 'NOC', 'Games', 'Ye
ar', 'Season', 'City', 'Sport', 'Event', 'Medal']

Info sur les données :
<class 'pandas.DataFrame'>
RangeIndex: 271116 entries, 0 to 271115
Data columns (total 15 columns):
 #   Column  Non-Null Count   Dtype
---  ------  --------------   -----
 0   ID      271116 non-null  int64
 1   Name    271116 non-null  str
 2   Sex     271116 non-null  str
 3   Age     261642 non-null  float64
 4   Height  210945 non-null  float64
 5   Weight  208241 non-null  float64
 6   Team    271116 non-null  str
 7   NOC     271116 non-null  str
 8   Games   271116 non-null  str
 9   Year    271116 non-null  int64
 10  Season  271116 non-null  str
 11  City    271116 non-null  str
 12  Sport   271116 non-null  str
 13  Event   271116 non-null  str
 14  Medal   39783 non-null   str
dtypes: float64(3), int64(2), str(10)
memory usage: 31.0 MB
None
```

In [2]:
```
df_2016 = df[df['Year'] == 2016].copy()
df_2016_medals = df_2016[df_2016['Medal'].notna()].copy()
```

```
print(f"Nombre de médailles en 2016 : {len(df_2016_medals)}")
print(f"Nombre de sports : {df_2016_medals['Sport'].nunique()}")
print("\n", df_2016_medals.head())
```

```
Nombre de médailles en 2016 : 2023
Nombre de sports : 34

       ID                        Name Sex   Age  Height  Weight        Team
\
158   62            Giovanni Abagnale   M  21.0   198.0    90.0       Italy
161   65            Patimat Abakarova   F  21.0   165.0    49.0  Azerbaijan
175   73                    Luc Abalo   M  31.0   182.0    86.0      France
450  250         Saeid Morad Abdevali   M  26.0   170.0    80.0        Iran
794  455  Denis Mikhaylovich Ablyazin   M  24.0   161.0    62.0      Russia

     NOC        Games  Year  Season            City       Sport  \
158  ITA  2016 Summer  2016  Summer  Rio de Janeiro      Rowing
161  AZE  2016 Summer  2016  Summer  Rio de Janeiro   Taekwondo
175  FRA  2016 Summer  2016  Summer  Rio de Janeiro    Handball
450  IRI  2016 Summer  2016  Summer  Rio de Janeiro   Wrestling
794  RUS  2016 Summer  2016  Summer  Rio de Janeiro  Gymnastics

                                       Event   Medal
158                Rowing Men's Coxless Pairs  Bronze
161               Taekwondo Women's Flyweight  Bronze
175                   Handball Men's Handball  Silver
450  Wrestling Men's Middleweight, Greco-Roman  Bronze
794           Gymnastics Men's Team All-Around  Silver
```

In [3]:
```
sport_medals = df_2016_medals['Sport'].value_counts()
top_5_sports = sport_medals.head(5)

print("Top 5 sports avec le plus de médailles en 2016 :")
print(top_5_sports)

# Filtrer le dataset pour ne garder que les top 5 sports
df_top5 = df_2016_medals[df_2016_medals['Sport'].isin(top_5_sports.index)].c
print(f"\nNombre de médailles dans les top 5 sports : {len(df_top5)}")
```

```
Top 5 sports avec le plus de médailles en 2016 :
Sport
Athletics    192
Swimming     191
Rowing       144
Football     106
Hockey        99
Name: count, dtype: int64

Nombre de médailles dans les top 5 sports : 732
```

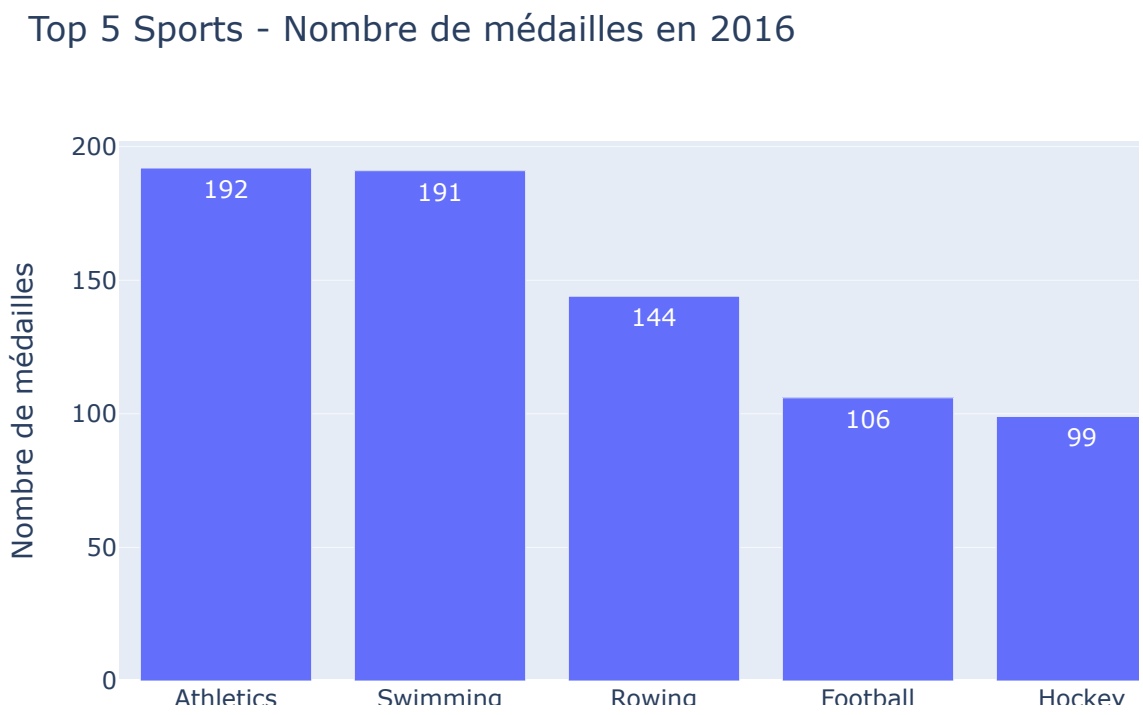
In [ ]:
```
# VISUALISATION 1 : Bar plot - Nombre de médailles par sport (top 5)
print("\n=== VISUALISATION 1 : Médailles par sport ===")

import plotly.express as px

fig1 = px.bar(
    x=top_5_sports.index,
```

```
    y=top_5_sports.values,
    labels={'x': 'Sport', 'y': 'Nombre de médailles'},
    title='Top 5 Sports - Nombre de médailles en 2016',
    text=top_5_sports.values
)
fig1.update_traces(textposition='auto')
fig1.show()
```

```
VISUALISATION 1 : Médailles par sport
```

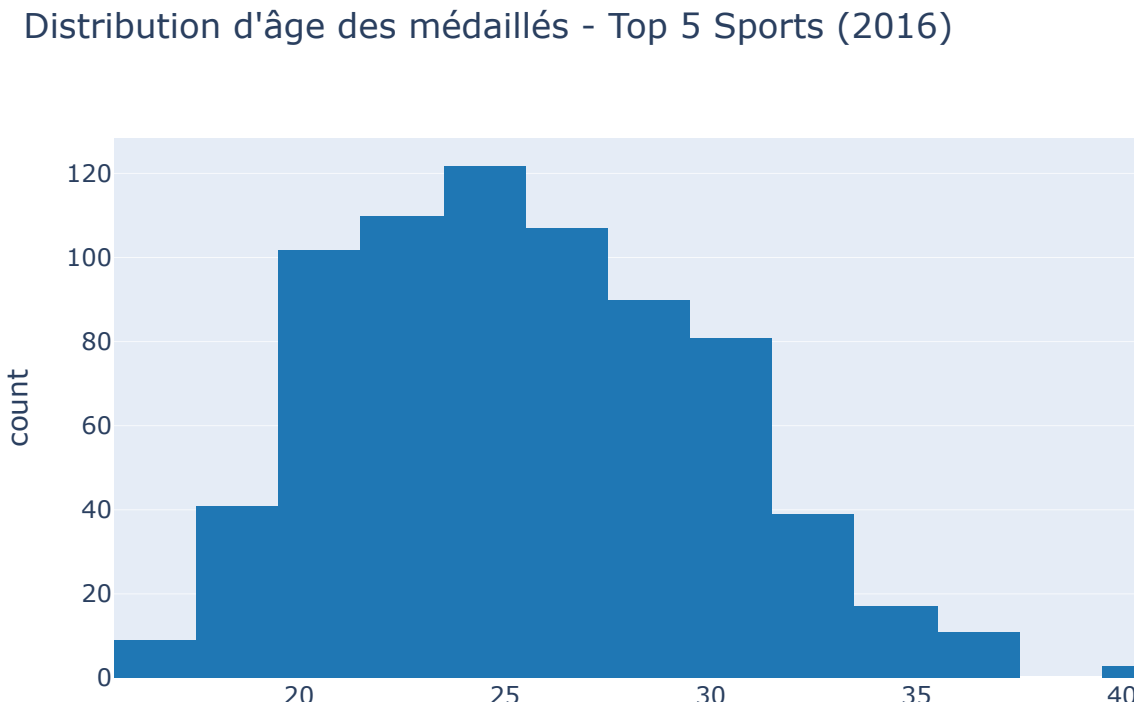


In [ ]:
```
# VISUALISATION 2 : Histogram - Distribution d'âge des médaillés (top 5 spor
print("\n=== VISUALISATION 2 : Distribution d'âge ===")

# Vérifier les valeurs manquantes d'âge
print(f"Valeurs manquantes d'âge : {df_top5['Age'].isna().sum()}")

fig2 = px.histogram(
    df_top5,
    x='Age',
    nbins=20,
    title='Distribution d\'âge des médaillés - Top 5 Sports (2016)',
    labels={'Age': 'Âge', 'count': 'Nombre de médaillés'},
    color_discrete_sequence=['#1f77b4']
)
fig2.show()
```

```
VISUALISATION 2 : Distribution d'âge
Valeurs manquantes d'âge : 0
```

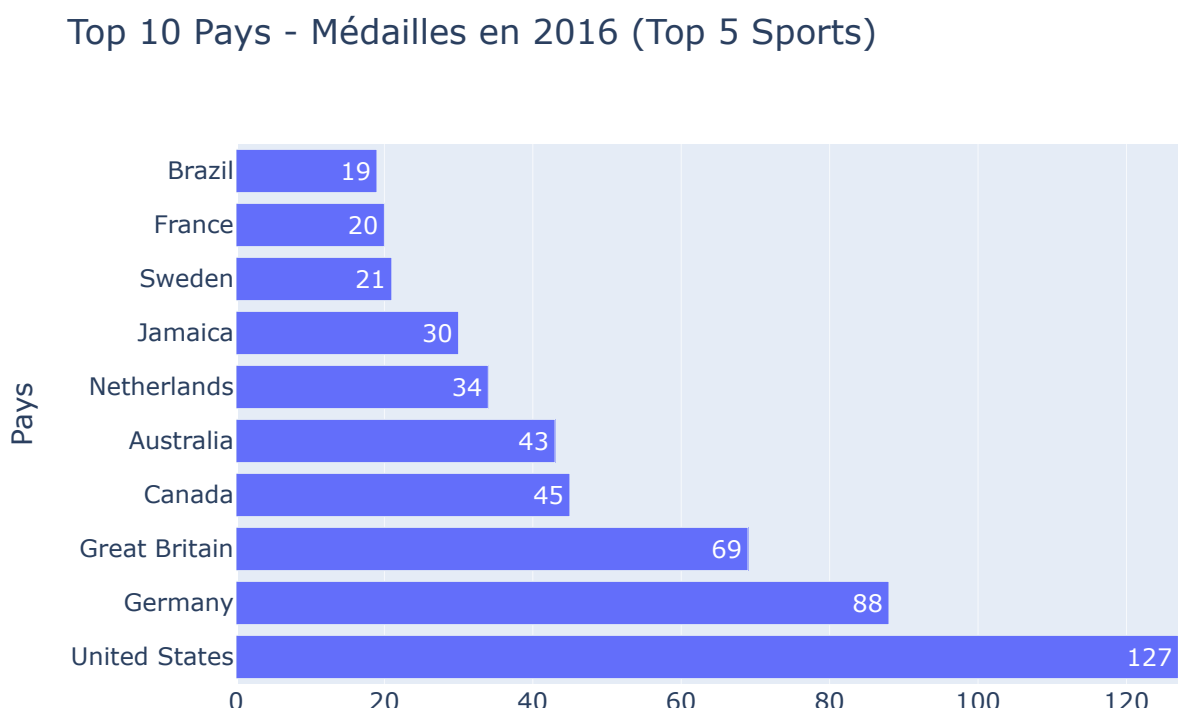


In [ ]:
```
# VISUALISATION 3 : Bar plot – Médailles par équipe nationale (top 10 pays)
print("\n=== VISUALISATION 3 : Médailles par pays ===")

# Compter les médailles par pays
country_medals = df_top5['Team'].value_counts().head(10)

fig3 = px.bar(
    x=country_medals.values,
    y=country_medals.index,
    orientation='h',
    title='Top 10 Pays - Médailles en 2016 (Top 5 Sports)',
    labels={'x': 'Nombre de médailles', 'y': 'Pays'},
    text=country_medals.values
)
fig3.update_traces(textposition='auto')
fig3.show()
```

```
VISUALISATION 3 : Médailles par équipe nationale
```

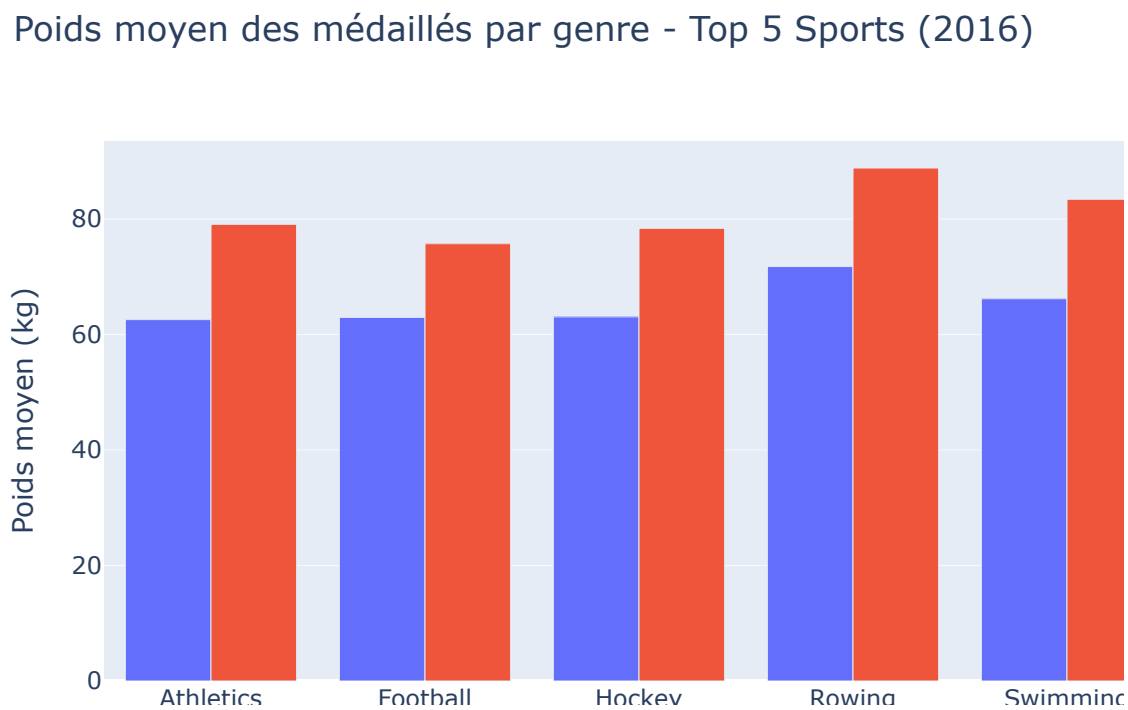


In [ ]:
```
# VISUALISATION 4 : Bar plot – Poids moyen par genre et sport (top 5)
print("\n=== VISUALISATION 4 : Poids moyen par genre ===")

# Vérifier les valeurs manquantes de poids
print(f"Valeurs manquantes de poids : {df_top5['Weight'].isna().sum()}")

# Calculer le poids moyen par sport et genre
weight_by_gender_sport = df_top5.dropna(subset=['Weight']).groupby(['Sport',
weight_by_gender_sport.columns = ['Sport', 'Genre', 'Poids moyen']

print("\nPoids moyen par sport et genre :")
print(weight_by_gender_sport)

fig4 = px.bar(
    weight_by_gender_sport,
    x='Sport',
    y='Poids moyen',
    color='Genre',
    title='Poids moyen des médaillés par genre - Top 5 Sports (2016)',
    labels={'Poids moyen': 'Poids moyen (kg)', 'Genre': 'Genre'},
    barmode='group'
)
fig4.show()
```

```
VISUALISATION 4 : Poids moyen par genre
Valeurs manquantes de poids : 5

Poids moyen par sport et genre :
       Sport Genre  Poids moyen
0  Athletics     F    62.578947
1  Athletics     M    79.074468
2   Football     F    62.962963
3   Football     M    75.725490
4     Hockey     F    63.061224
5     Hockey     M    78.380000
6     Rowing     F    71.766667
7     Rowing     M    88.833333
8   Swimming     F    66.195876
9   Swimming     M    83.408602
```